DD2437 Lab Assignment 1a: Classification with a Single-Layer Perceptron

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1 Main objectives and scope of the assignment

The primary objective of this assignment is to investigate the capabilities and limitations of single-layer perceptrons (Threshold Logic Units) for binary classification tasks. Specifically, the study compares the classical Perceptron learning rule with the Delta (Widrow-Hoff) rule, evaluating their convergence dynamics, sensitivity to hyperparameters, and behavior in both sequential (online) and batch learning modes.

Furthermore, the scope includes analyzing the geometric limitations of linear classifiers. This is achieved by evaluating the network's performance on datasets that are not linearly separable (overlapping and bimodal distributions), investigating the impact of the bias weight on the decision boundary, and observing how sampling bias (via data subsampling) skews the network's generalization.

2 Methods

2.1 Data Generation and Representation

Datasets were generated using 2D multivariate normal distributions. For linearly separable data, 100 samples per class were generated with means $\mu_A = [1.0, 0.5]$ and $\mu_B = [-1.0, 0.0]$, using a standard deviation of $\sigma = 0.3$. For overlapping (non-linearly separable) data, the standard deviation was increased to $\sigma = 0.5$. A third dataset featured a bimodal distribution for Class A. To enable the network to learn an offset decision boundary, the input matrix X was augmented with a row of ones. Target labels T were set to $\{0, 1\}$ for the Perceptron rule and converted to $\{-1, 1\}$ for the Delta rule to ensure correct gradient direction.

2.2 Learning Algorithms

Three algorithmic variants were implemented from scratch using NumPy:

- **Perceptron Rule (Sequential):** Updates weights sample-by-sample based on the error computed *after* the step activation function ($\Delta W = \eta(t - y)x^T$).
- **Delta Rule (Sequential):** Updates weights sample-by-sample using the continuous error measured *before* thresholding ($\Delta W = \eta(t - Wx)x^T$).
- **Delta Rule (Batch):** Computes the error across the entire dataset simultaneously. To allow direct comparison using the same learning rate ($\eta = 0.01$), the accumulated gradient was normalized by the number of samples N ($\Delta W = \frac{\eta}{N}(T - WX)X^T$).

Performance was tracked per epoch using Mean Squared Error (MSE) and the misclassification rate. Weights were initialized from a zero-mean normal distribution.

3 Results and discussion

3.1 Classification with a single-layer perceptron

3.1.1 Algorithm Comparison and Convergence

The algorithms were first evaluated on the linearly separable dataset. As shown in Figure 1, both the Delta Sequential and Delta Batch algorithms successfully converged to a misclassification rate of zero, though their convergence dynamics differed significantly. The sequential (online) Delta rule converged almost instantaneously, dropping to zero misclassifications within the first 2 epochs, accompanied by a sharp, immediate plunge in Mean Squared Error (MSE). In contrast, the Delta Batch algorithm required approximately 130 epochs to reach zero misclassifications. Its MSE decayed smoothly and monotonically over the 200-epoch training period, reflecting a stable but much slower descent along the error surface resulting from the averaged batch gradient updates. Random weight initialization affected the exact starting error but did not prevent either variant from finding a valid separating hyperplane.

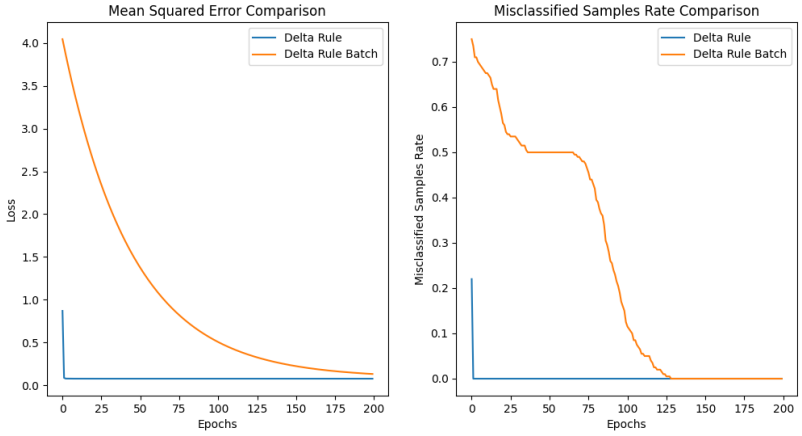
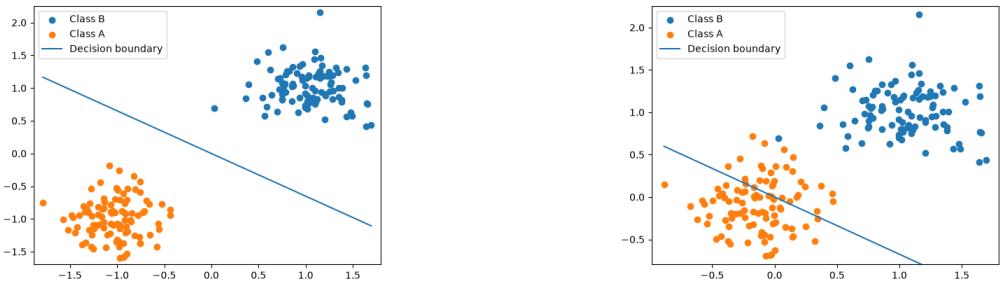


Figure 1: Learning curves comparing the Delta Rule (Sequential) and Delta Rule (Batch) on linearly separable data. Both variants successfully reduce the misclassification rate to zero.

3.1.2 Impact of Removing the Bias

When the bias term was removed ($w_0 = 0$), the decision boundary was geometrically constrained to pass exactly through the origin ($w_1x_1 + w_2x_2 = 0$). Experimental verification confirmed the hypothesis: the bias-free perceptron only converged when the distribution means μ_A and μ_B were located in strictly opposite quadrants (e.g., upper-right and lower-left), as shown in Figure 2a. When both means were placed in the same quadrant, the network failed to separate the data entirely, as a line anchored at the origin could not physically pass between the clusters, as shown in Figure 2b.



(a) Point clouds forced in the upper-right and the lower-left quadrants.

(b) Class A point cloud partially in the upper-right quadrant.

Figure 2: Impact of removing the bias and constraining the decision boundary.

3.2 Classification of data that are not linearly separable

3.2.1 Overlapping Distributions

For the dataset with increased variance ($\sigma = 0.5$), the classes overlapped, making perfect linear separation impossible. Figure 3 demonstrates the failure mode of the single-layer perceptron. Both the classical Perceptron and the Delta Rule failed to reach zero misclassifications, heavily oscillating or plateauing at a minimum error rate of approximately 2%. Notably, the Delta rule achieved a more stable MSE, as it optimizes for the continuous distance to the target rather than solely focusing on hard misclassifications.

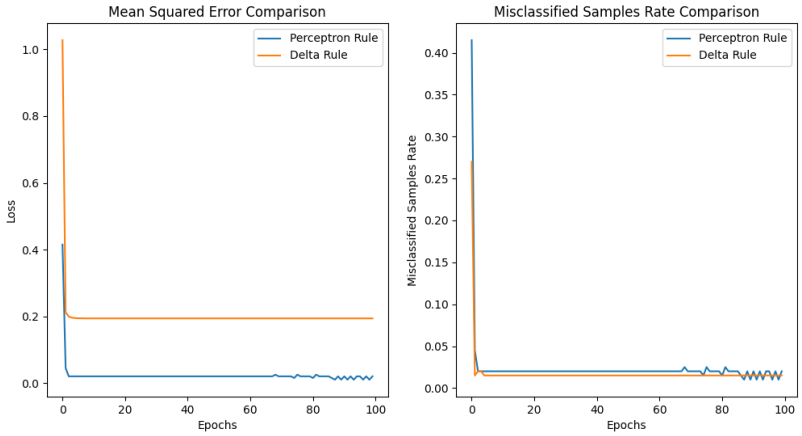


Figure 3: Learning curves for Perceptron vs. Delta Rule on overlapping, non-linearly separable data. Neither algorithm reaches zero misclassifications.

3.2.2 Bimodal Data and Sampling Bias

The Delta Batch algorithm was subsequently tested on a non-linearly separable bimodal dataset under four distinct subsampling conditions to evaluate the impact of sampling bias. Because an overall accuracy metric obscures performance disparities in unbalanced datasets, the misclassification rate was computed and tracked independently for Class A and Class B across three independent runs.

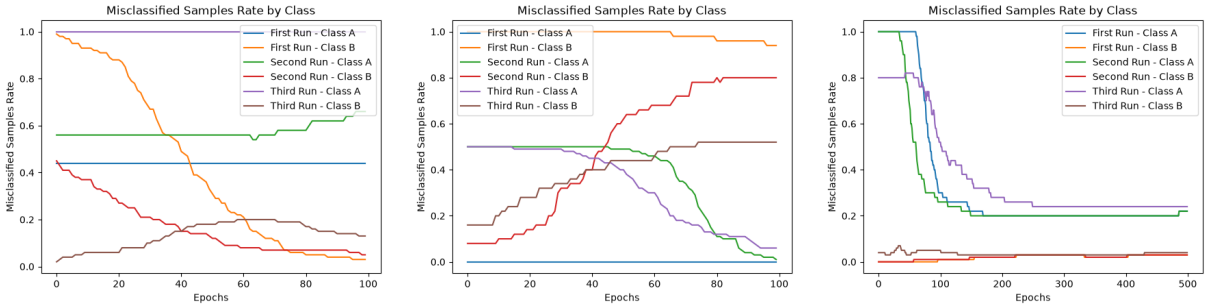
The experiments demonstrate that the single-layer perceptron's decision boundary is highly sensitive to both the frequency and spatial distribution of the training data (see Figure 4):

- **25% reduction in both classes:** The relative frequency balance between the classes is maintained. The decision boundary remains stable, yielding relatively symmetric and low

misclassification rates for both classes.

- **50% reduction in Class A (Figure 4a):** The introduced class imbalance causes the network to prioritize minimizing errors on the majority class (Class B). The decision boundary shifts deeply into Class A's region. As observed in the training logs, Class A's misclassification rate degrades severely, diverging to 100% in some runs depending on initialization, while Class B's error rate drops to near 0%.
- **50% reduction in Class B (Figure 4b):** The inverse effect is observed. The network favors the new majority (Class A), causing Class B's misclassification rate to spike (plateauing between 50% and 100% across different runs), while Class A achieves near-perfect classification.
- **Asymmetric spatial subsampling (Figure 4c):** By removing 20% of Class A where $x_1 < 0$ and 80% where $x_1 > 0$, the bimodal geometry of Class A is heavily depleted in its rightmost cluster. The decision boundary shifts to adapt to this new spatial reality, drawing a dividing line between Class B and the rest of Class A, which is concentrated on the left. The network converges to a stable but asymmetric error state.

Overall, these scenarios mathematically verify that a linear classifier's generalization is strictly bounded by its training distribution. Uniform or spatial data depletion physically skews the decision boundary, artificially inflating accuracy for the majority or spatially dominant class at the direct expense of the minority class.



(a) 50% of Class A removed. (b) 50% of Class B removed. (c) Asymmetric removal (Class A).

Figure 4: Per-class misclassification rates for the Delta Rule (Batch) across three independent runs under varying sampling bias conditions. Removing samples from one class physically rotates the decision boundary, artificially improving accuracy for the majority class while severely degrading the performance of the depleted minority class.

4 Final remarks

This lab demonstrated the fundamental mechanics and theoretical limitations of the single-layer perceptron. While both the Perceptron and Delta rules successfully classify linearly separable data, the Delta rule's optimization of continuous error yields smoother gradients and greater stability. The implementation of batch learning highlighted the necessity of gradient normalization to prevent exploding weights and revealed its susceptibility to local minima in asymmetric, non-separable datasets. Finally, the subsampling experiments emphasized that a model's generalization is only as robust as the representation of its training data; sampling bias physically rotates the decision boundary, artificially inflating accuracy for majority classes at the expense of minority classes.